GeoCrop: Revised Evaluation Metrics

Focused on performance and computation

F26-155-GeoCrop — Response to proposal panel comments P2–P4 — Measured on AOI1, October 2026

Each metric has a unit, a target and the value measured so far. Accuracy is kept only as a quality gate. **Status:** *met* = measured and within target; *to measure* = planned for Iteration 2.

1. Map and rendering performance (primary)

Metric	Unit	Target	Measured
Data per screen	KB	$<$ 1 MB	**337 KB** (prototype 43.3 MB)
Cold tile latency	ms, median / p95	$<$ 300 / $<$ 500	**47 / 54**
Cached tile latency	ms, median	$<$ 50	**1**
Time to first visible layer	s	$<$ 1	to measure
Frame rate while panning and zooming	FPS	$\geq$ 30	to measure
Layers visible at once without lag	count	$\geq$ 8	to measure
Objects drawn by the browser	count	minimise	24 tiles (prototype 101,435 polygons)

2. Processing and computation cost

Metric	Unit	Target	Measured
Upload-to-map time (AOI-sized image)	s	$<$ 120	**57**
Compute time per stage	s	reported	COG 0.7; cropland 3.8; crop type 10
Peak RAM per stage	MB	$<$ 4 GB (laptop)	357 / 395 / **3,145**; LightGBM 549
Inference cost	ms per 1,000 px	$<$ 100	LightGBM **35.8** (Presto 2,551)
Model size	MB	tens of MB or less	LightGBM **1.98** (RF 1,570)
Storage per result	KB	minimise	222 KB (prototype 43.3 MB)
Idle platform RAM (7 services)	GB	$<$ 2	**about 1.0**

3. Scalability and integration

Metric	Unit	Target	Status
Data per screen as images grow (1, 5, 10 AOIs)	KB	roughly flat	to measure
Concurrent tile requests (about 20 users)	p95 ms	$<$ 500	to measure (load test)
Time to add a new model	files changed	manifest only	**met** (TC-03)
Parallel jobs without blocking the map	count	$\geq$ 2	to measure

4. Quality gate (secondary)

Metric	Target	Measured
Accuracy / macro-F1 on spatial test blocks	with 95% interval	AOI1 76.1% / 0.686; AOI2 92.2% / 0.803
Accuracy loss from the lighter model	$\leq$ 1 point vs. best	LightGBM 0.748 vs. RF 0.752

How the metrics are used

- **Cost per unit of result:** ms per 1,000 pixels, KB per screen, seconds per AOI, so the prototype and the platform are directly comparable.
- **Old against new:** the 130$\times$ data reduction and 1,021 ms JSON parse against 47 ms tiles are reported side by side.
- **Median and p95,** not only averages, to reflect real use.
- **Resource analysis:** time, peak RAM and storage are measured for every workflow stage and repeated for each new model.